

Deep Learning: Advanced Models and Methods (DL26) • G23

Aligning Small LLMs with GRPO for Strict JSON Generation

Rule-Based Reward Optimization for Schema-Conformant Output

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 github.com/GiuseppeBellamacina/GRPO-strict-generation

Small LLMs (<2B) Cannot Reliably Produce Valid JSON

- ✗ **31–39% Pass@1** for SmolLM2-135M
(39% No-Think, 31% with Think prompting)
- ✗ Unclosed brackets, type violations, conversational filler mixed with output
- **Goal:** use GRPO with rule-based rewards to reach >90% Pass@1 — no neural reward model
- **Dense, deterministic, interpretable** signal drives alignment on a single GPU

✗ Typical failure

```
Sure! Here's the JSON you asked for:  
{  "name": "Alice",  "age": 30,  
  "skills": ["Python", "ML" }  
// Missing ] and extra text
```

✓ Expected output

```
“`json {\"name\": \"Alice\", \"age\": 30,  
  \"skills\": [\"Python\", \"ML\"]}\"`“
```

5 Models (135M–2B), Synthetic Dataset, Single GPU

Model	Params	Arch
SmolLM2-135M-Instruct	135M	LLaMA-like
SmolLM2-360M-Instruct	360M	LLaMA-like
Qwen2.5-0.5B-Instruct	0.5B	Qwen2.5
TinyLlama-1.1B-Chat	1.1B	LLaMA 2
Gemma-2-2B-it	2B	Gemma 2

Dataset

-  24 parametric templates (8/difficulty)
-  4500 training samples
-  300 eval (100 S / 100 M / 100 H)
-  No ground-truth JSON — reward evaluates directly

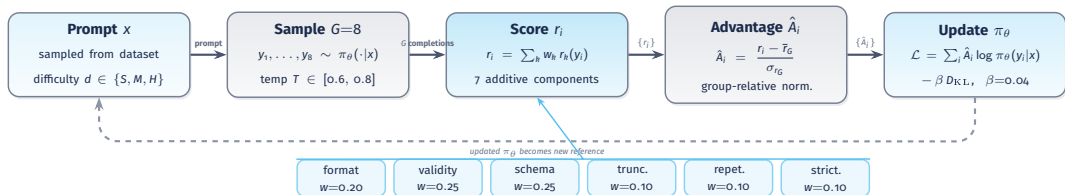
Configuration

-  4-bit NF4 quantization
-  LoRA $r=16$, $\alpha=32$
-  1× NVIDIA L40S (48 GB)
-  2500 steps / model
-  Paged AdamW 8-bit
-  LR 5×10^{-6} , cosine

4 strategies tested:

Think / No-Think × Standard / Curriculum
Identical hyperparameters across all models

GRPO: Group-Relative Advantage Eliminates the Critic



Key idea: No critic network needed. Advantages are computed *within* each group of $G=8$ completions. Additive reward composition ensures gradient is always non-zero — avoids **zero-advantage collapse**.

7 Additive Rewards + 3-Stage Curriculum Prevent Collapse

Component	Range	W	Measures
 format	[0, 1]	.20	“ ‘ json “ ‘ fence
 validity	[0, 1]	.25	JSON parseability
 schema	[0, 1]	.25	Keys, types, nesting
 truncation	[−1, 1]	.10	Unclosed braces
 repetition	[−1, 1]	.10	Token loops
 strictness	[−1, 1]	.10	Extra text outside JSON
 reasoning	[−0.5, 1]	.25 [*]	<think> quality

^{*}Think mode only; weight redistributed when off



Curriculum = safety net

Prevents regression in strong models.

Gemma-2-2B: −1.67pp (standard) → +0.67pp (curriculum)

Graduated partial credit

Binary validity gives no gradient for near-valid JSON. Position-based scoring (0.20–0.70) keeps learning alive.

All 5 Models Converge to 87–99% Pass@1 After 2 500 Steps

Model	Pre	Peak	Best Configuration	Δ
SmolLM2-135M	31%	90.00%	Think / Standard	+58.67 pp
SmolLM2-360M	78%	96.67%	No-Think / Standard	+19.00 pp
Qwen2.5-0.5B	92%	98.00%	Think / Curriculum	+6.33 pp
TinyLlama-1.1B	79%	99.33%	No-Think / Curriculum	+20.00 pp
Gemma-2-2B	96%	97.67%	Think / Curriculum	+1.33 pp

All 5 models, best configuration per model. Pre = same-config baseline (Think lowers small-model baselines by ~ 8 pp).

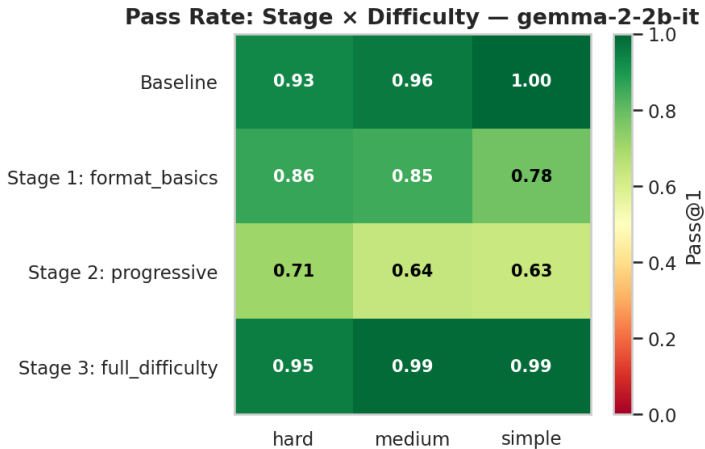
+58.67 pp largest gain

SmolLM2-135M: 31% $\rightarrow$ 90%

99.33% best overall

TinyLlama-1.1B (No-Think / Curriculum)

Gemma-2-2B Temporarily Collapses: Curriculum Pacing Must Match Capacity



Gemma-2-2B • Think / Curriculum • Pass@1 per stage & difficulty

U-shaped collapse

96% → 83% → 66% → 98%

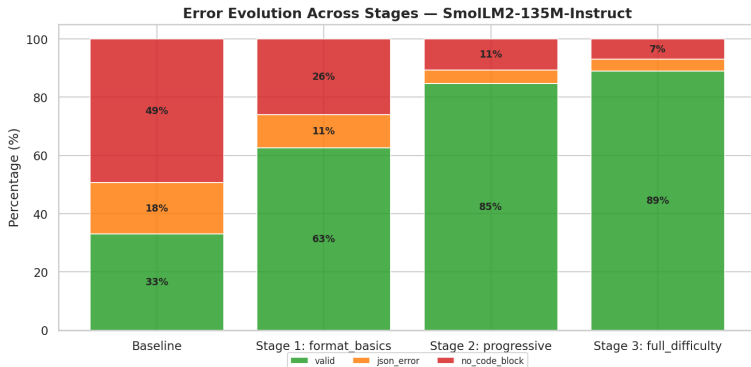
Forcing `<think>` on trivial schemas confuses an already-strong model.

Full recovery only once hard schemas dominate (Stage 3).

Lesson

Strong models need less warm-up. Standard training causes -1.67pp regression; Curriculum recovers to $+0.67\text{pp}$.

Errors Vanish Stage by Stage: Format First, Then Structure



SmolLM2-135M • No-Think / Curriculum • Error types per stage

Stage 1

Dominant error:
`no_code_block`
Model learns the
“`‘json’`” fence first

Stage 2

Format mastered.
Residual: `json_error`
Focus shifts to structure

Stage 3

Near-zero errors.
Only edge-case
schema failures remain

Reward Design Matters More Than Model Size

Main Message

GRPO with 7 additive rule-based rewards brings **all 5 models** (135M–2B) to **87–99% Pass@1** on strict JSON generation, single GPU, identical hyperparameters.

Takeaways

1. **Standard** optimal for weak models (135M: +58.67pp)
2. **Curriculum** safety net for strong models (prevents Gemma regression)
3. **Think mode** <2pp marginal gain (small models can't produce <think>)

Limitations

- JSON-only — not directly generalizable
- Fully synthetic dataset; real-world gap unknown
- Fixed hyperparameters across models

Future Work

- Extend to YAML, XML, SQL output
- Real-world API-call datasets
- Function reward ablation comparison
- Per-model hyperparameter tuning



Think vs. No-Think

Small models (135M, 360M, 0.5B) never generate `<think>` tokens; only Gemma and TinyLlama can. 135M peaks 90% in Think by exploiting other rewards, not reasoning.



Curriculum = Safety Net

Produces absolute highest scores (TinyLlama 99.33%, Qwen 98.00%). Standard causes Gemma to regress.



Inverted-U Gain Pattern

Smallest models gain most: 135M +58pp. Mid-range: +13–26pp. Already-strong: +0–8pp (ceiling).



Hard Tier Drives Improvement

Baselines: 17–20% hard pass for 135M. Post-GRPO: 86–99% on hard prompts. Simple saturates to 100% in Stage 1.



Backup Slides

Additional detail for Q&A

Baseline Pass@1 — Per-Difficulty Breakdown

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	39.00%	58.00%	39.00%	20.00%
SmolLM2-360M	77.67%	85.00%	86.00%	62.00%
Qwen2.5-0.5B	92.67%	95.00%	95.00%	88.00%
TinyLlama-1.1B	79.33%	89.00%	76.00%	73.00%
Gemma-2-2B	98.33%	100%	100%	95.00%

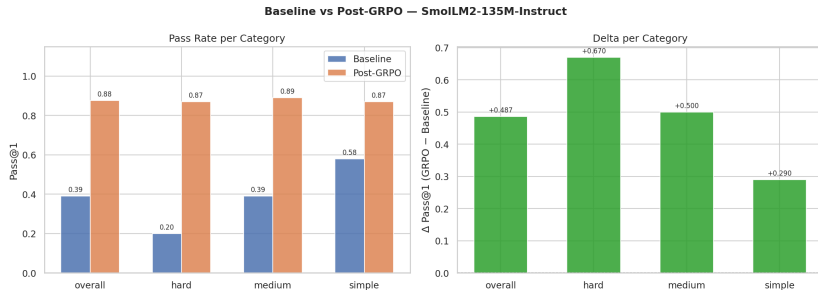
300-sample balanced eval (100 S / 100 M / 100 H). Config: No-Think / Standard.

Reward System — Full Think/No-Think Weight Table

Component	Range	Think	NoThink	What it measures
 format	[0, 1]	0.10	0.20	“ ‘ json “ ‘ fence → 1.0 generic fence → 0.5 none → 0.0
 validity	[0, 1]	0.15	0.25	JSON parseability; partial credit by error position (0.20–0.70)
 schema	[0, 1]	0.20	0.25	Keys, types, nesting depth, required fields
 reasoning	[−0.5, 1]	0.25	0.0	<think> block quality; penalises missing/copied
 truncation	[−1, 1]	0.10	0.10	Unclosed braces / unterminated strings
 repetition	[−1, 1]	0.10	0.10	Token loops, repeated lines, duplicate blocks
 strictness	[−1, 1]	0.10	0.10	Extra text outside JSON >50% of output

Design principle: Additive composition, no hard gating. Gradient is always non-zero; avoids [zero-advantage collapse](#). When Think is off, reasoning weight is redistributed proportionally.

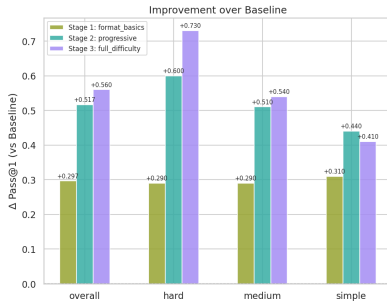
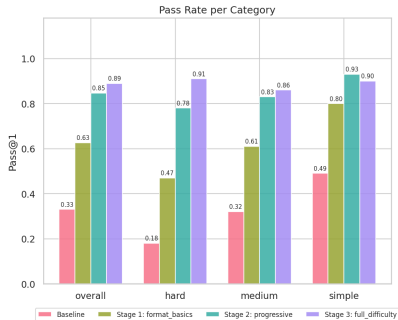
SmolLM2-135M — No-Think / Standard Comparison



SmolLM2-135M • No-Think / Standard • Pass@1 by difficulty tier, baseline vs. GRPO

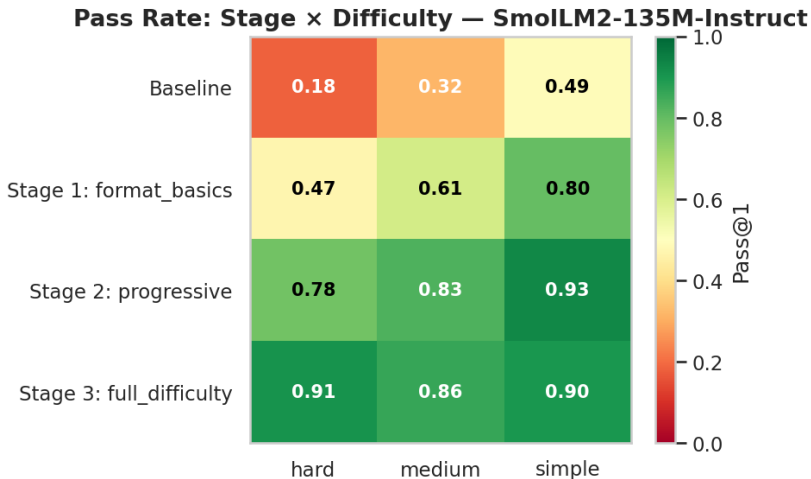
SmolLM2-135M — Curriculum Progression (No-Think)

Curriculum Progression — SmolLM2-135M-Instruct



SmolLM2-135M • No-Think / Curriculum • 33% → 62.7% → 84.7% → 89.0%

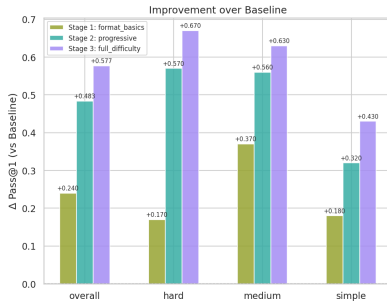
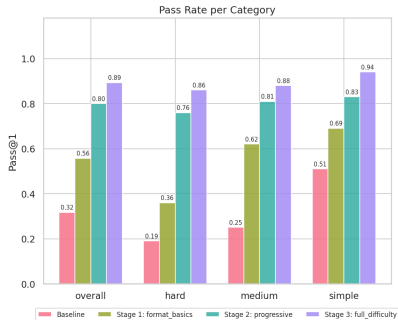
SmolLM2-135M — Difficulty Heatmap (No-Think / Curriculum)



SmolLM2-135M • No-Think / Curriculum • Pass@1 per difficulty tier at each stage

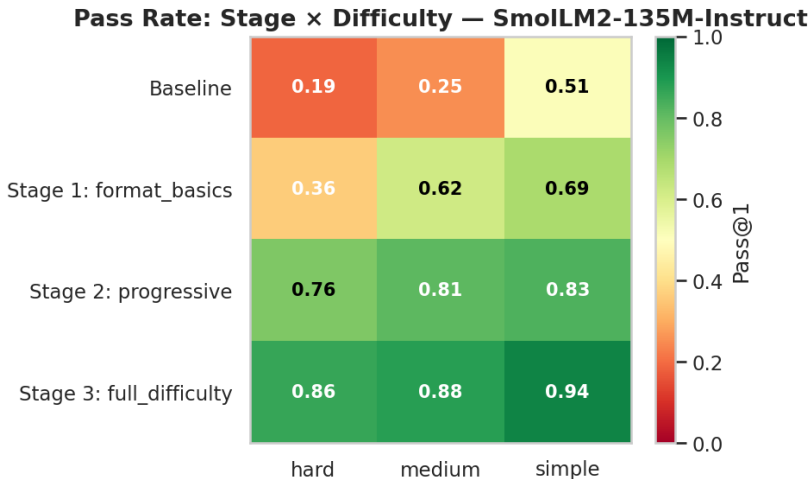
SmolLM2-135M — Curriculum Progression (Think)

Curriculum Progression — SmolLM2-135M-Instruct



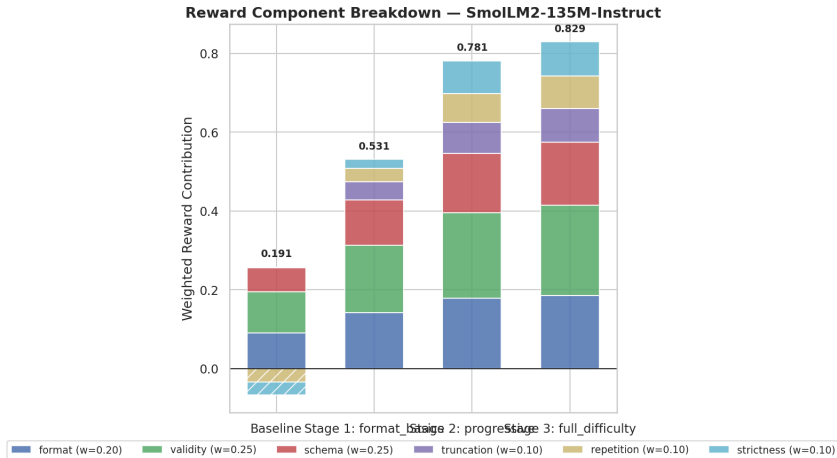
SmolLM2-135M • Think / Curriculum • 31.7% → 55.7% → 80.0% → 89.3%

SmolLM2-135M — Difficulty Heatmap (Think / Curriculum)



SmolLM2-135M • Think / Curriculum • Pass@1 per difficulty tier at each stage

Reward Component Contributions at Final Checkpoint



SmoLM2-135M, No-Think / Curriculum